

Joel Teply

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Dear Hiring Team,

I am applying for DFA's senior engineering role supporting production AI-enabled platforms. I am a Kansas native, a University of Kansas Computer Engineering graduate, and a hands-on engineer with more than twenty years of software experience, including extensive work in machine learning, computer vision, high-performance computing, and production AI systems.

I have worked with machine learning as an engineering discipline for years, particularly where models have to operate inside real products rather than isolated experiments. At Cambrian, I designed and built computer vision and ML systems for spatial computing, augmented reality, and perception. That work included CNN-based semantic segmentation and surface estimation, pose and geometry reconstruction, synthetic training data generated in Unreal Engine, temporal and multi-frame inference, GPU-accelerated image processing, and real-time deployment on mobile-class hardware.

Those systems had demanding constraints. Multiple neural networks could be performing expensive inference asynchronously while the application still needed to maintain a coherent understanding of the scene at frame rate. I built architectures combining learned inference with tracking, optical flow, structural geometry, RANSAC-based estimation, temporal interpolation, and multi-frame state accumulation. The objective was never simply model accuracy—it was building an intelligent system that remained fast, stable, and useful under real-world conditions.

I co-founded Cambrian and served first as CEO and later CTO while remaining deeply hands-on technically. Our computer vision technology was integrated into enterprise products, including systems that allowed customers to visualize spaces transformed with paint and flooring in real time or through still imagery. Delivering that technology required considerably more than ML models: production APIs, AWS infrastructure, S3-backed data and ML pipelines, Docker CPU/GPU deployments,

automated testing, scalable inference services, SDKs, native applications, browser software, and extensive performance engineering.

My current work on Continuum pushes further into modern AI systems. I built a distributed platform for persistent AI agents that combines local model inference, evaluation, scheduling, continual learning, LoRA-based adaptation, persistent memory and state, peer-to-peer communication, and heterogeneous CPU/GPU execution. Much of my recent work has focused on the systems surrounding models: how inference is scheduled, how learning and evaluation interact, how state persists, how workloads move across hardware, and how AI